

Genetic report for longevity

Immunity and Immune Ageing

Inflammation, immunosenescence, vaccine response, susceptibility to infections
and autoimmune overlap

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With age, the immune system is at the same time more inflammaging. IL-6, TNF, CRP) and less effective (immunosenescence: fewer native T cells, worse response to vaccines and infections). This report reads the genes of cytokines (IL6, IL6R, TNF, IL1B, IL10, CRP), the scores of CRP and leukocyte counts, the response to vaccines and the susceptibility to infections (MBL2, IFITM3, IFNL4, OAS1, 3p21.31, herpes zoster) and the overlap with autoimmunity (STAT4, PTPN22, IRF5, HLA).

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

28

variants

23

in the chip

3

imputed
(DR2 $\geq$ 0.8)

2

not found

20

Scores with
percentile

Overall reading. Immune genetics is, in general, favourable. In inflammation, CRP is low (12, 26 and 26 percentiles in three scores) and IL-6 in the medium (46 and 50); IL6R has 1 allele Ala358, which slightly lowers the classical IL-6 signal. Against: IL6 -174 GG (weak evidence) and higher CRP alleles in HNF1A and NLRP3. In immunosenescence, neutrophils are on the low side (the 20th percentile), lymphocytes in 70 and leukocytes in 64, and do not have the haplotype JAK2 46/1 of clonal hematopoiesis. In antiviral defense and vaccines: HLA-DPB1 AA (good response to hepatitis B vaccine), IFNL4 CC (best viral elimination), 1 protective copy of OAS1 and without severe COVID-19 allele of 3p21.31 (severe COVID score at the 27th percentile); high genetic vitamin D (the 91st percentile). In infections, MBL2, TLR2 and TLR4 normal; sepsis at the 23rd percentile; pneumonia at percentile 84.2 by the complete genomic score (the simple count of 25 loci of this report gives 38); herpes zoster is at the 75th percentile, the limit of unfavourable. In autoimmunity, there is no risk alleles in STAT4, PTPN22, IRF5, CTLA4, HLA-B27 or in the shared epitope of HLA-DRB1; rheumatoid arthritis at the 19th percentile and multiple sclerosis in 24. The lupus at the 79th percentile is the only signal against, with partial coverage of HLA (weak).

Favourable

- **CRP (71 444 variants):** the 12th percentile
- **Neutrophils (492 loci):** the 20th percentile
- **Vitamin D (25-OH-D) in blood (62 loci):** the 91st percentile
- **Rheumatoid arthritis (256 variants):** the 19th percentile
- **Multiple sclerosis (127 loci):** the 24th percentile
- **Sepsis:** the 23rd percentile
- **JAK2 rs10974944 CC:** Haplotype JAK2 46/1: the G allele increases 3–4 $\times$ the probability of acquiring the JAK2 V617F mutation (clonal haematopoiesis and myeloproliferative neoplasms).
- **DARC/ACKR1 rs2814778 TT:** Duffy-null: the C allele gives genetically low neutrophils (without clinical significance); if absent, neutrophil counts are interpreted by the usual values.
- **HLA-DPB1 rs9277535 AA:** HLA-DPB1 (3'UTR): allele A is associated with better elimination of hepatitis B and a better response to the vaccine; G = lower response.
- **IFNL4 rs12979860 CC:** IFNL4/IL28B: the genotype CC is associated with spontaneous elimination and response to treatment of hepatitis C; T = minor elimination.
- **TLR2 rs5743708 GG:** R753Q: Allele A reduces TLR2 signaling (Gram-positive bacteria, mycobacteria).
- **IRF5 rs2004640 GG:** IRF5: T allele increases the production of interferon type I and the risk of lupus.
- **CTLA4 rs231775 AA:** CTLA4 +49A>G: G allele reduces T lymphocytes' brake (thyroiditis, diabetes type 1).

Unfavourable

- **Systemic lupus erythematosus (57 loci):** the 79th percentile
- **IL6 rs1800795 GG:** -174G>C: the G allele was associated with more IL-6 after stimuli and, in a study in men, the least probability of reaching centenary; the effect was not confirmed in large studies.
- **HNF1A rs2244608 AA:** HNF1A: liver transcription factor; CRP locus.
- **NLRP3 rs12239046 CC:** NLRP3: inflammasome.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 23 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Inflammaging: cytokines and CRP

IL-6, TNF- α and IL-1 β rise with age and predict frailty, cardiovascular disease and mortality; IL-10 stops inflammation. C-reactive protein (CRP) is the routine marker. The IL-6 receptor (IL6R Asp358Ala) is a target validated by genetics: the Allele Low the classical signal and coronary disease.

Ageing mechanisms: Inflammaging; Chronic Inflammation; Intercellular communication alteration; Vascular Aging

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
CRP (77 loci)	Typical / intermediate	the 26th percentile · z=-0.59	Higher score = CRP genetically higher. PGS000314 (PGS Catalog; GWAS-meaning (lead and secondary) variants; C-reactive protein; PGP000092, PMID 32527150). Variants with genotype: 76 of 77 (covering of 99.4% coefficients).
CRP (71 444 variants)	Favourable	the 12th percentile · z=-1.16	Higher score = CRP. PGS001946 (PGS Catalog; Penalized regression; C-reactive protein; PGP000263, PMID 34995502). Variants with genotype: 68830 of 71444 (covering of 96.4% coefficients).
IL-6 in blood (552 426 variants)	Typical / intermediate	the 46th percentile · z=-0.08	Higher score = IL-6. PGS000252 (PGS Catalog; Bayesian method; Interleukin-6 (IL-6) serum levels; PGP000089, PMID 33067605). Variants with genotype: 541930 of 552426 (covering of 94.6% coefficients).
IL-6 (loci of GWAS)	Typical / intermediate	the 50th percentile · z=0.13	Higher score = IL-6. Loci score of GWAS Catalog (p $\leq$ 510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 9 of 10 loci with genotype. Studies: Ahluwalia TS (Interleukin-6 levels); Folkersen L (Interleukin-6 levels); Kanoni S (Interleukin-6 levels in non-alcoholic fatty liver disease x mastiha supplementation interaction); Meeks KAC (Interleukin-6 levels); Sherman BT (Interleukin-6 levels in HIV infection). Variants with genotype: 9 of 10 (covering of 93.3% coefficients).
TNF-α (GWAS loci)	SNP not found in the data	—	Higher score = higher TNF- α .
CRP (PGS002873, 1.1 million variants)	Typical / intermediate	the 26th percentile	Health Report score (Bayesian method). Source: Health Report, imputed genome.

Favourable

- CRP (71 444 variants) (the 12th percentile)

Unfavourable

- IL6 rs1800795 GG
- HNF1A rs2244608 AA
- NLRP3 rs12239046 CC

Unfavourable variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
IL6	rs1800795 7:22727026 · C/G/T	GG	chip	G (2; 58%)	weak	-174G>C: G allele was associated with more IL-6 after stimuli and, in a study in men, the least probability of reaching centenary; the effect was not confirmed in large studies.
HNF1A	rs2244608 12:120979185 · A/G	AA	chip	A (2; 66%)	moderate	HNF1A: liver transcription factor; CRP locus. Meaning: GWAS Catalog: C-reactive protein levels (Sinnott-Armstrong N 2021; p=0e+00); G ↓ allele
NLRP3	rs12239046 1:247438293 · T/A/C/G	CC	chip	C (2; 65%)	moderate	NLRP3: inflammasome. Meaning: GWAS Catalog: C-reactive protein levels (Dehghan A 2011; p=1e-15); allele C ↑

Typical / intermediate variants (5)

Gene	Variant	Genotype	Source	Evidence	Interpretation
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Unfavourable allele (copying; frequency in reference)						
IL6R	rs2228145 1:154454494 · A/C/T	CA	chip	A (1; 64%)	strong	Asp358Ala: C allele (358Ala) increases soluble IL-6R, low classical IL-6 signal, CRP and coronary artery disease (similar to a mild tocilizumab). A = Asp.
TNF	rs1800629 6:31575254 · G/A	GG	chip	A (0; 13%)	weak	-308G>A: allele A was associated with more TNF-α and the risk of several inflammatory diseases in small studies.
IL10	rs1800896 1:206773552 · T/C	CT	chip	T (1; 55%)	weak	-1082A>G: G allele is associated with more IL-10 (anti-inflammatory) and was more frequent in centenarians; T (A in the gene) : low producer.
CRP	rs1205 1:159712443 · C/T	TC	chip	C (1; 69%)	moderate	3'UTR of CRP: low T allele to basal CRP (~20%); C = higher CRP. Mendelian randomization shows that CRP itself does not cause coronary disease.
IL1RN	rs6743376 2:113074756 · C/A	CA	chip	A (1; 69%)	moderate	IL1RN (IL-1 receptor antagonist): CRP locus. Meaning: GWAS Catalog: C-reactive protein levels (MTAG) (Koskeridis F 2022; p=4e-30); allele A ↑

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
IL1B	rs16944 2:112837290 · A/G	AG	chip	—	weak	-511C>T: inconsistent associations with IL-1β. Information.
APOC1	rs4420638 19:44919689 · A/G	GA	imputation DR2=1.00	A (1; 80%)	moderate	APOE/APOC1: the G allele, linked to ε4, low to CRP (paradox effect, without benefit); here APOE is ε3/ε3. No judgment. Meaning: GWAS Catalog: C-reactive protein levels (Sinnott-Armstrong N 2021; p=0e+00); G ↓ allele

Immunosenescence and blood cells

The counts of lymphocytes and neutrophils are very hereditary. Low lymphocytes and high neutrophils (high N/L ratio) accompany immunosenescence and inflammation. The haplotype JAK2 46/1 predisposes to clonal hematopoiesis with a mutation JAK2.

Ageing mechanisms: Immunosenescence; Thymic Involution; Loss of T Naive cells; Restriction of the TCR repertoire; Clonal hematopoiesis (CHIP)

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Total leukocytes (636 loci)	Typical / intermediate	the 64th percentile · z=0.35	Higher score = more leukocytes (associated with more inflammation and mortality). PGS000191 (PGS Catalog; Genome-wide signifiers; White blood cell count; PGP000078, PMID 32888494). Variants with genotype: 513 of 636 (covering of 81.7% coefficients).
Lymphocytes (621 loci)	Typical / intermediate	the 70th percentile · z=0.47	Higher score = more lymphocytes (low side accompanies immunosenescence). PGS000172 (PGS Catalog; Genome-wide signifiers; Lymphocyte count; PGP000078, PMID 32888494). Variants with genotype: 497 of 621 (covering of 80.4% coefficients).
Neutrophils (492 loci)	Favourable	the 20th percentile · z=-0.88	Higher score = more neutrophils. PGS000182 (PGS Catalog; Genome-wide signifiers; Neutrophil count; PGP000078, PMID 32888494). Variants with genotype: 388 of 492 (covering of 79.7% coefficients).

Favourable

- Neutrophils (492 loci) (the 20th percentile)
- JAK2 rs10974944 CC
- DARC/ACKR1 rs2814778 TT

Unfavourable

None.

Favourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
JAK2	rs10974944 · 9:5070831 · C/G	CC	imputation · DR2=1.00	G (0; 27%)	moderate	Haplotype JAK2 46/1: the G allele increases 3-4× the probability of acquiring the JAK2 V617F mutation (clonal haematopoiesis and myeloproliferative neoplasms).
DARC/ACKR1	rs2814778 · 1:159204893 · T/C	TT	chip	C (0; 1%)	strong	Duffy-null: the C allele gives genetically low neutrophils (without clinical significance); if absent, neutrophil counts are interpreted by the usual values.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
BCL2	rs187971642 · 18:63213468 · T/C	TT	chip	C (0; 0%)	moderate	BCL2: lymphocyte survival. Meaning: GWAS Catalog: Lymphocyte count (Chen MH 2020; p=1e-21); allele T ↑

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CD247	— · —	—	SNP not found in the data	—	moderate	CD247 (CD3ζ): T cell receptor signal; lymphocyte locus. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Response to vaccines and antiviral defense

The response to vaccines (hepatitis B, influenza) depends on HLA class II (HLA-DPB1, HLA-DQ) and antiviral innate immunity (IFITM3, IFNL4, OAS1). Vitamin D modulates innate immunity.

Ageing mechanisms: Immunosenescence; Loss of T Naive cells; Intercellular Communication

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Vitamin D (25-OH-D) in blood (62 loci)	Favourable	the 91st percentile · $z=1.32$	Higher score = vitamin D genetically higher. PGS019960 (PGS Catalog; genome-wide signifier SNPs; Serum 25-Hydroxyvitamin D; PGP000849, PMID 39053885). Variants with genotype: 54 of 62 (covering of 87.9% coefficients).

Favourable

- Vitamin D (25-OH-D) in blood (62 loci) (the 91st percentile)
- HLA-DPB1 rs9277535 AA
- IFNL4 rs12979860 CC

Unfavourable

None.

Favourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
HLA-DPB1	rs9277535 6:33087084 · A/G	AA	chip	G (0; 27%)	moderate	HLA-DPB1 (3'UTR): allele A is associated with better elimination of hepatitis B and a better response to the vaccine; G = lower response.
IFNL4	rs12979860 19:39248147 · C/G/T	CC	chip	T (0; 31%)	strong	IFNL4/IL28B: the genotype CC is associated with spontaneous elimination and response to treatment of hepatitis C; T = minor elimination.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
OAS1	rs10774671 12:112919388 · G/A/C	GA	chip	A (1; 65%)	moderate	OAS1 (splicing): the G allele (p46 isoform, of neanderthal origin) protects from severe COVID-19 A = less protection.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
IFITM3	rs12252 11:320772 · A/G	—	SNP not found in the data (Imputed with $DR2=0.78 < 0.8$)	C (—; 0%)	moderate	IFITM3 (splicing): C allele was associated with severe influenza; it is rare in the reference population.

Susceptibility to infections

MBL2 (mannose binding lectin, opsonization), TLR4 (bacterial recognition), 3p21.31 (LZTFL1, COVID-19 severe) and herpes zoster scores, low respiratory infections, cellulitis and seropositivity for Epstein-Barr virus.

Ageing mechanisms: Immunosenescence; Chronic Inflammation; Dysbiosis

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Herpes zoster (81 variants)	Typical / intermediate	the 75th percentile · z=0.71	Higher score : higher risk. PGS001131 (PGS Catalog; penalized regression; Zoster [herpes zoster] (time-to-event); PGP000244, PMID 35324888). Variants with genotype: 62 of 81 (covering of 66.3% coefficients).
Acute low respiratory infection (400 variants)	Typical / intermediate	the 45th percentile · z=-0.09	Higher score : higher risk. PGS000925 (PGS Catalog; penalized regression; Unspecified below respiratory infection (time-to-event); PGP000244, PMID 35324888). Variants with genotype: 339 of 400 (covering of 84.4% coefficients).
Cellulitis and superficial abscesses (131 variants)	Typical / intermediate	the 67th percentile · z=0.45	Higher score : higher risk. PGS001869 (PGS Catalog; Penalized regression; Surface cellulitis and abscess; PGP000263, PMID 34995502). Variants with genotype: 128 of 131 (covering of 98.9% coefficients).
Seropositivity for Epstein-Barr virus (35 variants)	Typical / intermediate	the 70th percentile · z=0.52	Higher score = more probability of antibodies (especially HLA class II). PGS001036 (PGS Catalog; penalized regression; Epstein Barr virus nuclear antigen 1 IgG seropositivity; PGP000244, PMID 35324888). Variants with genotype: 27 of 35 (covering of 68.9% coefficients).
Pneumon	Unfavourable (genomic score)	Loci count: the 38th percentile · z=-0.21; complete genomic score: percentile 84.2	The complete genomic score (Pneumococcus report) prevails over this simple count. Higher score : higher risk. Few loci. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 20 of 25 loci with genotype. Studies: Butler-Laporte G (Pneumonia (90-day mortality)); Field AI (Pneumonia); Gelemanovic A (Pneumonia); Halligan NLN (Pneumonia); Reay WR (Pneumonia); Sakaue S (Pneumonia); Verma A (Pneumonia (PheCode 480)); Verma A (Pneumonia due to fungi (mycoses) (PheCode 480.3)). Variants with genotype: 20 of 25 (covering of 80.0% coefficients).
Sepsis	Favourable	the 23rd percentile · z=-0.95	Higher score : higher risk. Few loci. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 21 of 36 loci with genotype. Studies: Butler-Laporte G (Sepsis (28-day mortality)); Butler-Laporte G (Sepsis (90-day mortality)); Butler-Laporte G (Sepsis (hospital admission)); Douville NJ (Sepsis-associated acute kidney injury) Douville NJ (Sepsis-associated acute kidney injury (model 3)); Guillen-Guide B (Acute respiratory distress syndrome in sepsis); Hernandez-Beeftink T (Survival in sepsis); Scherag A (Mortality in sepsis). Variants with genotype: 21 of 36 (covering of 58.3% coefficients).
COVID-19 grave	Typical / intermediate	the 27th percentile · z=-0.52	Higher score : higher risk of severe disease. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 58 of 67 loci with genotype. Studies: COVID-19 Host Genetics Initiative (COVID-19) It says Almeida S (COVID-19 (hospitalized covid vs population)); Gong B (COVID-19) Kosmicki JA (COVID-19 (hospitalized vs population)); Kousathanas A (COVID-19) Pairo-Castineira E (COVID-19) Pairo-Castineira E (COVID-19) Westerman KE (COVID-19 (severe vs population) x multiple deprivation index interaction (joint test)). Variants with genotype: 58 of 67 (covering of 86.4% coefficients).
Herpes labial (herpes simplex virus)	Typical / intermediate	the 53rd percentile · z=-0.21	Higher score : higher risk. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 2 of 4 loci with genotype. Studies: Butler-Laporte G (Anti-herpes simplex virus 2 IgG seropositivity); Butler-Laporte G (Herpes simplex virus 2 mgG-1 antibody levels). Variants with genotype: 2 of 4 (covering of 48.8% coefficients).

Favourable

- Sepsis (the 23rd percentile)
- TLR2 rs5743708 GG

Unfavourable

None.

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TLR2	rs5743708 4:153705165 · G/A	GG	chip	A (0; 2%)	moderate	R753Q: Allele A reduces TLR2 signaling (Gram-positive bacteria, mycobacteria).

Typical / intermediate variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MBL2	rs1800450 10:52771475 · C/T	CC	chip	T (0; 14%)	moderate	MBL2 G54D (B allele): T (A in the gene) very low to MBL; homozygotes have more infections in childhood and immunosuppression.
TLR4	rs4986790 9:117713024 · A/G/T	AA	chip	G (0; 6%)	weak	D299G: G allele reduces lipopolysaccharide response (less inflammation, plus some Gram-negative infections).
LZTFL1	rs35044562 3:45867532 · A/G	AA	imputation DR2=0.99	G (0; 8%)	strong	3p21.31: the G allele (Neanderthal haplotype) doubles the risk of severe COVID-19.

Autoimmune overlap

Many risk genes are shared by several autoimmune diseases: STAT4, PTPN22, IRF5, CTLA4, HLA-DRB1 (shared epitope) and HLA-B27. Rheumatoid arthritis scores, lupus, multiple sclerosis and 1 type diabetes.

Ageing mechanisms: Chronic Inflammation; Immunosenescence; Intercellular Communication

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Rheumatoid arthritis (256 variants)	Favourable	the 19th percentile · z=-0.97	Higher score : higher risk. PGS001875 (PGS Catalog; Penalized regression; Rheumatoid arthritis; PGP000263, PMID 34995502). Variants with genotype: 249 of 256 (covering of 99.3% coefficients).
Systemic lupus erythematosus (57 loci)	Unfavourable	the 79th percentile · z=0.78	Higher score : higher risk. Partial coverage (HLA). PGS000328 (PGS Catalog; Genomewide-meaning variants (sourced from PMID:28509669); Systemic lupus erythematosus; PGP000099, PMID 31826855). Variants with genotype: 24 of 57 (covering of 37.4% coefficients).
Multiple sclerosis (127 loci)	Favourable	the 24th percentile · z=-0.71	Higher score : higher risk. PGS000809 (PGS Catalog; Clumping of variants associated with multiple sclerosis; Multiple sclerosis PGP000194, PMID 34088990). Variants with genotype: 101 of 127 (covering of 84.4% coefficients).
1 type diabetes (33 loci)	Typical / intermediate	the 42nd percentile · z=-0.24	Higher score : higher risk. PGS000021 (PGS Catalog; SNP associations cured from the literature; Type 1 diabetes (T1D); PGP000011, PMID 26577414). Variants with genotype: 25 of 25 (covering of 100.0% coefficients).

Favourable

- Rheumatoid arthritis (256 variants) (the 19th percentile)
- Multiple sclerosis (127 loci) (the 24th percentile)
- IRF5 rs2004640 GG
- CTLA4 rs231775 AA

Unfavourable

- Systemic lupus erythematosus (57 loci) (the 79th percentile)

Favourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
IRF5	rs2004640 7:128938247 · T/G	GG	chip	T (0; 53%)	moderate	IRF5: T allele increases the production of interferon type I and the risk of lupus.
CTLA4	rs231775 2:203867991 · A/G/T	AA	chip	G (0; 36%)	moderate	CTLA4 +49A>G: G allele reduces T lymphocytes' brake (thyroiditis, diabetes type 1).

Typical / intermediate variants (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
STAT4	rs7574865 2:191099907 · T/A/G	GG	chip	T (0; 23%)	strong	STAT4 (3 intron): T allele increases the risk of lupus, rheumatoid arthritis and Sjögren syndrome.
PTPN22	rs2476601 1:113834946 · A/G/T	GG	chip	A (0; 9%)	strong	R620W: allele A increases the risk of several autoimmune diseases (rheumatoid arthritis, type 1 diabetes, thyroiditis, vitiligo).
HLA-DRB1		AA	chip	G (0; 22%)	moderate	

	rs660895 6:32609603 · A/G					Shared epitope of HLA-DRB1 (seropositive rheumatoid arthritis): the G allele marks the risk.
HLA-B27	rs4349859 6:31398010 · G/A/C	GG	chip	A (0; 5%)	strong	HLA-B*27 tag (ankylosing spondylitis, anterior uveitis): the A allele marks B27.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Intercellular Communication	IL6 GG; HNF1A AA; NLRP3 CC; Systemic lupus erythematosus (57 loci) (the 79th percentile)	CRP (71 444 variants) (the 12th percentile); Vitamin D (25-OH-D) in blood (62 loci) (the 91st percentile); HLA-DPB1 AA; IFNL4 CC; Rheumatoid arthritis (256 variants) (the 19th percentile); Multiple sclerosis (127 loci) (the 24th percentile); IRF5 GG; CTLA4 AA	CRP (77 loci) (the 26th percentile); IL-6 in blood (552 426 variants) (the 46th percentile); IL-6 (GWAS loci) (the 50th percentile); CRP (PGS002873, 1.1 million variants) (the 26th percentile); IL6R CA; CRP TC; IL1RN CA; OAS1 GA
Chronic Inflammation	IL6 GG; HNF1A AA; NLRP3 CC; Systemic lupus erythematosus (57 loci) (the 79th percentile)	CRP (71 444 variants) (the 12th percentile); Sepsis (the 23rd percentile); TLR2 GG; Rheumatoid arthritis (256 variants) (the 19th percentile); Multiple sclerosis (127 loci) (the 24th percentile); IRF5 GG; CTLA4 AA	CRP (77 loci) (the 26th percentile); IL-6 in blood (552 426 variants) (the 46th percentile); IL-6 (GWAS loci) (the 50th percentile); CRP (PGS002873, 1.1 million variants) (the 26th percentile); IL6R CA; CRP TC; IL1RN CA; Herpes zoster (81 variants) (the 75th percentile)
Vascular Aging	IL6 GG; HNF1A AA; NLRP3 CC	CRP (71 444 variants) (the 12th percentile)	CRP (77 loci) (the 26th percentile); IL-6 in blood (552 426 variants) (the 46th percentile); IL-6 (GWAS loci) (the 50th percentile); CRP (PGS002873, 1.1 million variants) (the 26th percentile); IL6R CA; CRP TC; IL1RN CA
Inflammaging	IL6 GG; HNF1A AA; NLRP3 CC	CRP (71 444 variants) (the 12th percentile)	CRP (77 loci) (the 26th percentile); IL-6 in blood (552 426 variants) (the 46th percentile); IL-6 (GWAS loci) (the 50th percentile); CRP (PGS002873, 1.1 million variants) (the 26th percentile); IL6R CA; CRP TC; IL1RN CA
Immunosenescence	Systemic lupus erythematosus (57 loci) (the 79th percentile)	Neutrophils (492 loci) (the 20th percentile); JAK2 CC; DARC/ACKR1 TT; Vitamin D (25-OH-D) in blood (62 loci) (the 91st percentile); HLA-DPB1 AA; IFNL4 CC; Sepsis (the 23rd percentile); TLR2 GG; Rheumatoid arthritis (256 variants) (the 19th percentile); Multiple sclerosis (127 loci) (the 24th percentile); IRF5 GG; CTLA4 AA	Total leukocytes (636 loci) (the 64th percentile); Lymphocytes (621 loci) (the 70th percentile); BCL2 TT; OAS1 GA; Herpes zoster (81 variants) (the 75th percentile); Acute low respiratory infection (400 variants) (the 45th percentile); Cellulitis and surface abscesses (131 variants) (the 67th percentile); Seropositivity for Epstein-Barr virus (35 variants) (the 70th percentile)
Dysbiosis	—	Sepsis (the 23rd percentile); TLR2 GG	Herpes zoster (81 variants) (the 75th percentile); Acute low respiratory infection (400 variants) (the 45th percentile); Cellulitis and surface abscesses (131 variants) (the 67th percentile); Seropositivity for Epstein-Barr virus (35 variants) (the 70th percentile); Pneumonia (percentile 84.2 in the genomic score; 38 in loci count; COVID-19 severe (the 27th percentile); Herpes labial (herpes simplex virus) (percentile 53); MBL2 CC
Clonal hematopoiesis (CHIP)	—	Neutrophils (492 loci) (the 20th percentile); JAK2 CC; DARC/ACKR1 TT	Total leukocytes (636 loci) (the 64th percentile); Lymphocytes (621 loci) (the 70th percentile); BCL2 TT
Thymic Involution	—	Neutrophils (492 loci) (the 20th percentile); JAK2 CC; DARC/ACKR1 TT	Total leukocytes (636 loci) (the 64th percentile); Lymphocytes (621 loci) (the 70th percentile); BCL2 TT
Loss of T Naive Cells	—	Neutrophils (492 loci) (the 20th percentile); JAK2 CC; DARC/ACKR1 TT; Vitamin D (25-OH-D) in blood (62 loci) (the 91st percentile); HLA-DPB1 AA; IFNL4 CC	Total leukocytes (636 loci) (the 64th percentile); Lymphocytes (621 loci) (the 70th percentile); BCL2 TT; OAS1 GA
Repertory TCR restriction	—	Neutrophils (492 loci) (the 20th percentile); JAK2 CC; DARC/ACKR1 TT	Total leukocytes (636 loci) (the 64th percentile); Lymphocytes (621 loci) (the 70th percentile); BCL2 TT

Practical part

Ideas for immunity, crossed with the genetic plan v2 and current medication (rapamycin 4 mg of 6 in 6 weeks, LDN 4.5 mg, isotretinoin, lithium, tadalafil, dapagliflozin, retatrutide, telmisartan, melatonin). The current stack remains as it is; everything is suggestion (not applied). Rapamycin in low and intermittent doses improved the response to influenza vaccine in elderly (Mannick 2014), but remains an immunomodulator: the schedule of vaccines should take into account.

1. Recombinant herpes zoster vaccine

Genetics: Herpes zoster p75; rapamycin (immunomodulation)

The recombinant vaccine (2 doses, 2–6 months apart) prevents more than 90% of cases and post-herpetic neuralgia; it is indicated from the 50 years or earlier with immunomodulation. Discuss with the doctor. It is an inactivated vaccine (safe with rapamycin).

V2 genetic plan and medication: Nova. • suggestion (not applied)

2. Vaccines in the middle of rapamycin cycle

Genetics: HLA-DPB1 AA; IFNL4 CC (good antiviral response)

Influences every year and reinforcement of COVID-19 according to recommendations; vaccination in the middle of the 6 cycle weeks (≥ 1 –2 weeks after taking) and waiting for 1–2 weeks before following taking. Live vaccines (yellow fever, measles) only with the doctor. If it has already been vaccinated for hepatitis B, an anti-HBs trial confirms the protection.

V2 genetic plan and medication: Coherent with the plan v2 (pushed rapamycin). • suggestion (not applied)

3. Monitoring inflammation in routine analysis

Genetics: Low genetic CRP (p12–26); IL6 GG (weak); HNF1A and NLRP3

CRP of high sensitivity and the neutrophil/lymphocyte ratio of the blood count, in the analyzes it already does; as the genetic CRP is low, a CRP persistently above 1 mg/L deserves to look for the cause (gengivals, intestine, liver). Ferritin <100 also avoids pro-inflammatory iron.

V2 genetic plan and medication: Coherent with the v2 plan (ferritin ≤ 100). • suggestion (not applied)

4. Vitamin D: confirm the value with 10 000 IU

Genetics: High genetic vitamin D (p91)

With 10 000 IU/day and high 25-OH-D genetics, 25-OH-D and calcium 1–2 $\times$ /year should be confirmed; the K2 MK-7 240 μ g is already in the stack.

V2 genetic plan and medication: Similar to the plan (D3 10 000 IU); it only adds confirmation in the analysis. • suggestion (not applied)

5. Omega-3 and fiber against inflammation

Genetics: Inflammaging (HNF1A, NLRP3, IL6)

Omega-3 3.2–4 g/day of EPA+DHA (fish oil is already in the stack) and fiber as high as possible (psyllium, PHGG, legumes, oats): both lower CRP and feed butyrate producers. Zero alcohol, as it already does.

V2 genetic plan and medication: Similar to the v2 plan (Omega-3 3.2–4 g; more fiber is always better). • suggestion (not applied)

6. Natural anti-inflammatory agents: only with normal ALT

Genetics: Inflammaging

Pausing curcumin, luteolin, apigenin and quercetin while the ALT is high is a suggestion (not applied; the real stack has not changed); it does not reintroduce them for immune reasons before that. Senolytic (fisetin + quercetin) also with normal ALT.

V2 genetic plan and medication: Similar to the v2 plan. • suggestion (not applied)

7. Sleep, exercise and zinc

Genetics: Immunosenescence (p20 neutrophils, p70 lymphocytes)

Short sleep reduces the response to vaccines (see Circadian Rhythm and Sleep); regular exercise maintains naive T cells. Zinc + copper, already in the stack, cover the function of the thymus.

V2 genetic plan and medication: The same as the plan. • suggestion (not applied)

8. Autoimmunity: no screening, only attention to signals

Genetics: STAT4, PTPN22, IRF5, CTLA4 without risk alleles; lupus p79 (partial coverage)

There is no indication for analysis of autoantibodies without symptoms. If there are joint pains with morning stiffness, new photosensitivity or wounds in the mouth: ANA and consultation. LDN, already taken, has a possibly favourable immunomodulatory effect.

V2 genetic plan and medication: Compatible with the plan. • suggestion (not applied)

Avoiding or with caution

- Vaccines alive after rapamycin. suggestion (not applied)
- Antibiotics without clear need (dysbiosis and resistance). suggestion (not applied)
- Reintroduce curcumin, luteolin, apigenin or quercetin with high ALT. suggestion (not applied)
- Sleeping little in the days around a vaccine. suggestion (not applied)
- Smoking and passive smoking (inflammation and respiratory infections). suggestion (not applied)

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. HLA is read by SNPs tag, not by typing; scores very dependent on HLA (lupus, diabetes type 1) have partial coverage.
- The variants of cytokines (IL6 -174, TNF -308, IL10 -1082, IL1B -511) come from studies of candidates with weak replication; the scores of CRP, IL-6 and leukocytes come from large GWAS.
- Immunosenescence depends much more on age, CMV, physical activity, sleep and medication (rapamycin, LDN) than these genes.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local statistical imputation (with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS000314 (PGS Catalog; GWAS-meaning (lead and secondary) variants; C-reactive protein; PGP000092, PMID 32527150 · PGS001946 (PGS Catalog; Penalized regression; C-reactive protein; PGP000263, PMID 34995502 · PGS000252 (PGS Catalog; Bayesian method; Interleukin-6 (IL-6) serum levels; PGP000089, PMID 33067605 · PGS000191 (PGS Catalog; Genome-wide signifiers; White blood cell count; PGP000078, PMID 32888494 · PGS000172 (PGS Catalog; Genome-wide signifiers; Lymphocyte count; PGP000078, PMID 32888494 · PGS000182 (PGS Catalog; Genome-wide signifiers; Neutrophil count; PGP000078, PMID 32888494 · PGS019960 (PGS Catalog; genome-wide signifier SNPs; Serum 25-Hydroxyvitamin D; PGP000849, PMID 39053885 · PGS001131 (PGS Catalog; penalized regression; Zoster [herpes zoster] (time-to-event); PGP000244, PMID 35324888 · PGS000925 (PGS Catalog; penalized regression; Unspecified below respiratory infection (time-to-event); PGP000244, PMID 35324888 · PGS001869 (PGS Catalog; Penalized regression; Surface cellulitis and abscess; PGP000263, PMID 34995502 · PGS001036 (PGS Catalog; penalized regression; Epstein Barr virus nuclear antigen 1 IgG seropositivity; PGP000244, PMID 35324888 · PGS001875 (PGS Catalog; Penalized regression; Rheumatoid arthritis; PGP000263, PMID 34995502 · PGS000328 (PGS Catalog; Genomewide-meaning variants (sourced from PMID:28509669); Systemic lupus erythematosus; PGP000099, PMID 31826855 · PGS000809 (PGS Catalog; Clumping of variants associated with multiple sclerosis; Multiple sclerosis PGP000194, PMID 34088990 · PGS000021 (PGS Catalog; SNP associations cured from the literature; Type 1 diabetes (T1D); PGP000011, PMID 26577414).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

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- GWAS Catalog and PGS Catalog (2026); Ensembl REST (public alleles and frequencies).

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